

MinVol terminal-handoff optimization audit

Repository source: `certificate/terminal_handoff_v11/README.md`

Result

This package records a bounded parameter audit of the public MinVol `v1.0.0` contact-geometric certificate. It does not introduce a new theorem architecture.

The audit found a genuine exact improvement. Split the terminal lower-branch row `03B02` at

$$R_m = \frac{611382837}{10^9} = 0.611382837$$

and retune the tangent radius independently on the two subintervals. The exact row bounds become

$$\begin{aligned} \text{O3B02A : } & \frac{131107875849897}{10^{15}} \pi = 0.411887539597799 \dots, \\ \text{O3B02B : } & \frac{131105572993743}{10^{15}} \pi = 0.411880304961823 \dots \end{aligned}$$

Both exceed the unchanged near-Jung branch

$$0.4118796712148324317327979810 \dots$$

Thus the near-Jung branch, rather than the lower radial partition, becomes the bottleneck. A conservative presentable theorem coefficient is

$$\frac{\text{Vol}(K)}{d^3} \geq \frac{26221074253\pi}{200000000000} = 0.4118796712122863775432315385 \dots$$

for every three-dimensional convex body of constant width d , conditional only on integrating these two new exact rows into the already released `v1.0.0` proof package. Its exact safety margins are:

- gain over public `v1.0.0`: more than `0.000002107431984103295052612615315...`;
- unchanged near-Jung branch over the candidate: more than `0.000000000002546054189566442454995...`;
- weakest split row over the candidate: more than `0.000000633749537026703427624968302...`.

The coefficient is deliberately rounded down far enough to preserve a useful cross-language numerical audit margin. The exact unchanged near-Jung rational lower is slightly larger.

Bounded search findings

Test	Exact outcome	Universal consequence
Local <code>s_0</code> retune, unsplit <code>03B02</code>	<code>0.411877563780308...</code>	Only a <code>6.28e-15</code> gain; negligible
Split <code>03B02</code> into 2 cells	minimum <code>0.411880304961823...</code>	Near-Jung becomes bottleneck
Split into 3 cells	minimum <code>0.411881218927344...</code>	No further universal gain
Split into 4 cells	minimum <code>0.411881675993877...</code>	No further universal gain
Double angular resolution, unsplit	<code>0.411878272868567...</code>	Improves <code>v1.0</code> but remains below near-Jung

Test	Exact outcome	Universal consequence
Double angular resolution, 2 cells	minimum 0.411881014196832...	No further universal gain
Shift R_* left by up to $4e-11$	near-Jung value decreases at every tested shift	No handoff improvement

The unchanged fan can move left only about $4.3639089872663e-11$ before its archived exact margin target is exhausted. Every tested left shift lowered the near-Jung branch while the retuned split lower branch stayed above it. Larger shifts would require changing the fan proof object and are outside this audit's parameter-only scope.

Proof authority and audits

- `terminal_handoff_v11_exact.py` is the proof authority for the two promoted split rows and the proposed universal comparison. It regenerates the radial chains and checks all accepted steps, adjacent grid extremality, cap-disjointness, tangent-gap sums, inherited-row comparisons, pi bounds, and the unchanged near-Jung rational lower using integer and rational arithmetic.
- `terminal_handoff_v11_exploration.py` replays the archived bounded search and reconstructs the near-Jung branch and left-shift behavior exactly from the public `v1.0.0` fan.
- `terminal_handoff_v11_binary64.py` is an independent numerical audit.
- `terminal_handoff_v11_audit.R` is a matching base-R numerical audit. It was prepared but was not executed in the construction environment because `Rscript` was unavailable.

The row generator was also regression-tested against the original immutable `03B02` arrays: it reproduced all 6,309 positive steps, all 18,562 antipodal steps, and the exact `v1.0.0` coefficient element-for-element. The transcript is archived under `transcripts/reference_generator_validation.txt`.

Replay

From this directory:

```
./run_all.sh
```

The exact proof objects can be regenerated deliberately with:

```
python3 terminal_handoff_v11_exact.py --write
```

Normal replay refuses stale or manually divergent JSON or arrays.

Claim boundary

This package establishes a **proposed exact v1.1.0 candidate**, not yet a new public release. Promotion still requires integration into the public repository, updated reconstruction and mutation tests, the base-R replay on a machine with R, checksum/document regeneration, and a green hosted workflow. The public `v1.0.0` tag and release should remain immutable.